

Electoral Cycle and Congressional Activity in Brazil
(1994-2022)

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1 Introduction

Holding politicians accountable in democracies heavily relies on the election rituals. Knowing this, the time leading up to an election is crucial for politicians to be as compelling as possible to the voters that they are deserving of reelection. In this sense, although officeholders’ full tenure in office should matter for voters’ evaluations of their performance, the voter effectively makes their decision as the election approaches. This creates an incentive for politicians to promote more “visible” policies closer to elections, as long as he is able to sustain a good reputation for the rest of the term.

This dynamic produced by the sharpness of elections in creating the need for voters to evaluate politicians is commonly called the “election cycle”. The literature on how election cycles reflect on budget allocation is vast. The overarching consensus is that government spending goes up in the year before an election because legislators want to provide goods to their constituencies and claim credit for them, boosting their reelection prospects (Ames 1977; Bueno 2023).

Text as data techniques allow us to look closer into not just aggregate measures of public spending, but systematically analyze what the whole of policy-making looks like within and across political cycles. We explore this possibility using Brazilian Congress bills between 1995 and 2022. The question we aim to explore is: Is the agenda of the Brazilian lower chamber in the year preceding an election different than the rest of the term?

2 Data & Methods

For this project, we use a dataset of the bills discussed on the floor of the Brazilian lower chamber between 1994 and 2022. We build upon primary data from the [Brazilian Congress API \(Proposições\)](#) and include covariates for the all coauthors’ and sponsors’ political party labels and the bill’s current status. We remove from the analyses bills proposed by the

President, the Senate (upper chamber), party leaderships, task forces, and civil society. Included in the analyses are thus all bills proposed by one or more lower chamber legislators (*Deputados*) and permanent and special committees, which are multipartisan by design.¹ We also exclude from the dataset observations that do not contain the bills content (variable *ementa* is empty).

Since we are interested in “business as usual” political cycles, we removed Dilma Rousseff’s second presidential term as her time in office ended in an impeachment process in 2016. We thus include 6 completed terms of 4 different presidents (exclude the period between 2015-2018). The table below illustrates the timeline:

President	FHC	FHC	Lula	Lula	Dilma	Bolsonaro
Term	1	2	1	2	1	1
Office	1995-1998	1999-2002	2003-2006	2007-2010	2011-2014	2018-2021
Party	PSDB	PSDB	PT	PT	PT	PSL/PL

Table 1: Timeline of Brazilian Presidents

We use Structural Topic Modeling to aggregate the bill contents into 30 topics based on their word contents. We started out trying a 30 topic specification because we believed it would be a parsimonious amount of topics, and would reveal to us most of the patterns we were interested in. Since a specification of 30 gave us substantively meaningful topics that we found interesting, we decided to keep it for our analysis.

We expect the content of the bill (*ementas*) to translate the agenda that different legislators want to put forward, and that we can thus identify variations along the electoral cycle. We have one main expectation: as elections approach, the legislative agenda changes. We think this should be particularly apparent when looking at topics that refer to public good and welfare provision like education, health, crime (which is very salient in Brazil), the

¹Deputados are assigned to committees based on outside specializations over certain policy topics.

environment, and unemployment/social security. We expect to see these topics being more prevalent in the data when we compare the agenda in the fourth year of a term against the other three years – since all members of Brazilian Congress are up for reelection every four years.

3 Results

Our model resulted in the following topic split, which we label below:

Table 2: Topics and Labels

Topic Number	High Frequency and Characteristic Words	Label
1	Human Rights Commission	humano, subcomissão, minorias, proferir, sere, redacao, estudar
2	Public Engagement and Debates	discutir, debat, realização, audiência, expositor, copa, cidades-sed
3	Legislative Amendments and Proposals	pronunciamento, diminuição, cabendo, ordem, implicação, aglutinativa, dia
4	Strategic Planning and Modifications	nova, navios-escolta, modifique-, modiffique-, estratégia, meta, anexo
5	Parliamentary Procedures and Regulations	deputado, câmara, regimento, sessão, solen, homenagem, ricd

6	Legal Proceedings and Justice	moção, repúdio, aplauso, pazuello, ataque, haiti, gsi
7	Executive Orders and Legislative Measures	provisória, suprimir, present, suprima-, altere-, dispositivo, subprojetosubatividad
8	Education Policies and Institutional Frameworks	ensino, estudant, fundament, currículo, curricular, curso, médio
9	Legislative Procedures and Voting Mechanisms	requerimento, despacho, aditamento, aposto, nomin, req, adiamento
10	Constitutional Amendments and Legal Frameworks	parágrafo, transitória, inclua-, couber, paragrafo, cessionário, cedent
11	Legislative Reviews and Technical Analyses	constitucionalidad, juridicidad, proferido, conclui, subemenda, psdb-sp, dem-sp
12	Legal Drafting and Amendments	alaltera, redação, liii, niova, mpv, dáa, adaptálo
13	Energy Policies and Environmental Regulations	aneel, renováveis, agraria, colonização, suden, aquicultura, ambient
14	Transportation Infrastructure and Government Projects	rodovia, br-, trecho, federativa, extraordinário, denomina, abr

15	Public Health Emergencies and Response Measures	covid-, pandemia, coronavírus, calamidad, vacinação, enquanto, reconhecido
16	Judicial Processes and Presidential Affairs	aviação, judiciais, ministra-chef, julgamento, licitatório, aeroportuária, reparação
17	National Security Measures and Legislative Decisions	munição, sinarm, incorporação, outra, cria, fogo, defin
18	Social Development Policies and Economic Strategies	comércio, organizado, inovação, ciência, climática, econômico, tecnologia
19	Presidential Engagements and Official Statements	pesar, srs, congratulaçõ, comparecerem, evangelização, convida, depor
20	Taxation Policies and Revenue Regulations	imposto, industrializado, ipi, bebida, microempresa, mercadoria, lucro
21	Legal Reforms and Legislative Changes	dispor, trânsito, permitir, prever, altera, contravençõ, auxílio-inclusão
22	Federal Regulations and Military Policies	carreira, bombeiro, plebiscito, distrito, auditoria-fisc, militar, gratificação

23	Municipal Suggestions and Ministerial Proposals	estiagem, impacta, samu, aed, aegypti, drenagem, uom
24	Legislative Approvals and Parliamentary Procedures	republic-sp, pl-am, prb-sp, pl-mg, portela, lincoln, psd-am
25	Ministerial Requests and State Reports	proger, informação, consignada, mantega, verba, técnicofinanceiro, aeronautica
26	Legislative Projects and Supplementary Proposals	apensação, desapensação, reconstituição, apensamento, desapensamento, desapensado, cabotagem
27	Public Administration and Service Reforms	administração, concurso, ps-gse, encaminha, processado, indireta, autarquia
28	Governmental Programs and Municipal Integration	dentro, aceleração, priorizaçõ, apa, alud, cadastrada, cadastramento
29	Broadcasting Regulations and Service Licensing	radiodifusão, comunitária, executar, exclusividad, dez, renova, outorgada
30	Regulatory Decrees and Administrative Actions	susta, contran, colegiada, normativa, ldo, elaboração, dotação

As we can see from the lists of topics and their most characteristic words, the vast majority of

topics seems to revolve somehow about the workings of the State, the process of lawmaking, the relationships to other branches of government, and public administration itself. Below we present a table with our assigned labels to each of the topics determined by the model, as printed above.

We expect the more “procedural” and governance-related types of legislation to not be as relevant for the common citizen, to whom services and good provision would be more consequential. We identify these policies to be gathered in Topic 8, which is about investments in education and school curriculum; Topic 13, which is about the environment and natural resource exploitation; Topic 15 is about health, but specifically about COVID health policy; Topic 18, which is about development, technology, and social security; and Topic 20, which is about employment, labor, and taxation. These will be our topics of interest for the rest of the analysis.

3.1 Test

We now move to testing whether our expectation that the topics selected above will be more prevalent in years before an election than in other term years. We will test this expectation’s face value from the frequency with which documents that fall into each of these topics in each year. We coded a bill year as the year in which a bill was first introduced in Congress, as we believe that this translates the idea of when that bill’s agenda is set. The years before elections in our sample are 1998, 2002, 2006, 2010, 2014, and 2018. If our expectations are correct, we should see peaks in frequency of documents belonging to topics 8, 13, 15, 18, and 20 in the years listed as pre-election years.

To calculate the prevalence of each topic across years, we need to manipulate the STM outputs and create a measure that reflects the idea of prevalence. To identify “how much” each document belongs to each of our topics of interest, we use the theta output from the STM model. We then aggregate the thetas of each topic of interest per year and divide this

sum by the number of bills proposed in that year, to see, relative to other years, whether that topic was more or less prevalent. We visualize our test by plotting each topic of interest's prevalence across years, and mark election years with dashed lines, which are the moments when we expect to see peaks in these topics' prevalence.

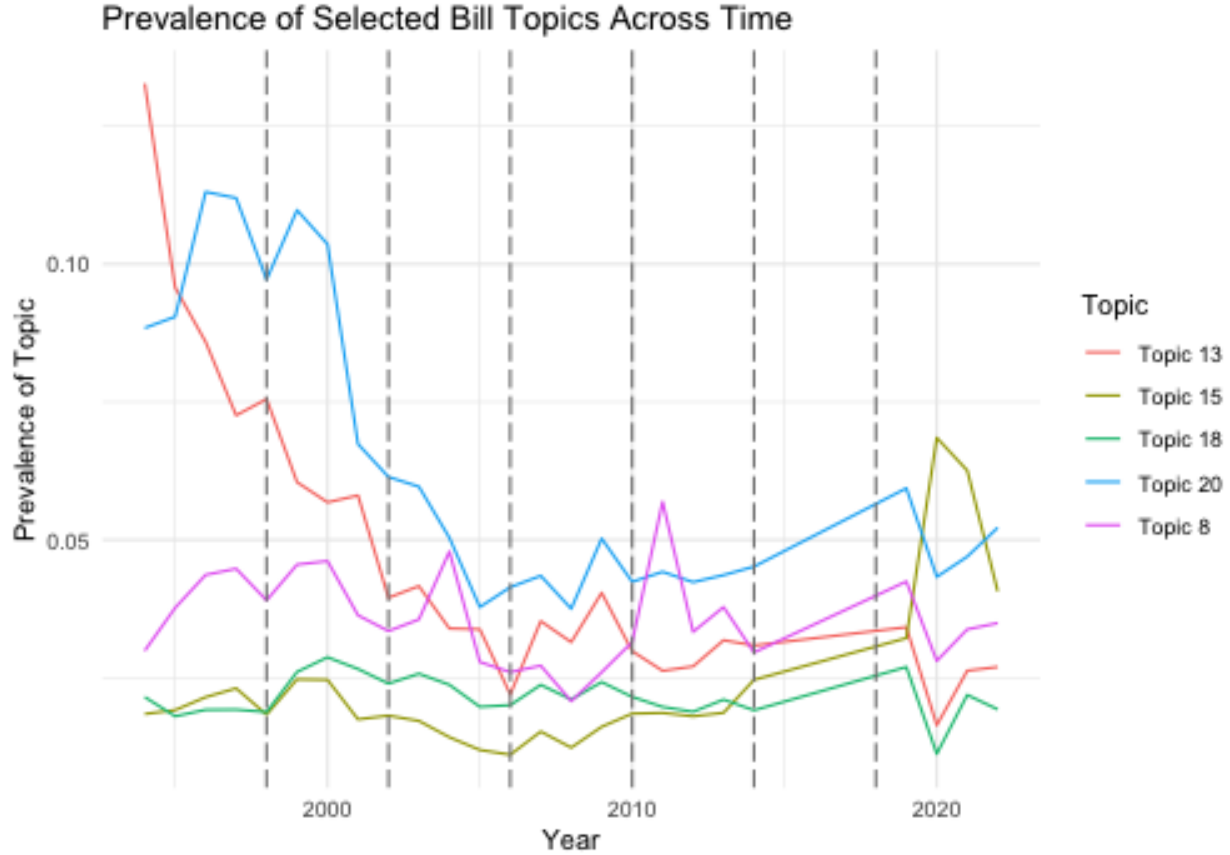


Figure 1: Bill Topic Prevalence Across Time

4 Discussion

We can see that our expectations do not find support in the data. We do not see peaks in the prevalence of bills within our topics of interest during election years. In fact, the graph generally shows the opposite of what was expected. We observed an overall trend that the prevalence of topics typically decreased in the time leading up the earlier elections (years

1998 to 2006). However, from 2010 onward, the decreasing prevalence trend changes to topics become slightly more prevalent leading up to the election years.

Looking at these topics of interest – (8) Education Policies and Institutional Frameworks, (13) Energy Policies and Environmental Regulations, (15) Public Health Emergencies and Response Measures, (18) Social Development Policies and Economic Strategies, and (20) Taxation Policies and Revenue Regulations – individually, there are very interesting trends observed. We saw a huge decrease in the prevalence of topic 13 from 1994 to 2022 which is not surprising, since environmental issues have gotten severe in Brazil, potentially as a consequence of deregulation by congress. One would expect bills regarding energy and environmental policies would increase during this time if Congress were more invested in regulating the environment, which could potentially improve environmental outcomes.

We also saw a significant decrease in topic 20 from 1994 to 2022, with a slight uptake in the middle around 2012. We can infer that this topic has become less popular within congress over time. Topic 8 generally fluctuated around the same prevalence level throughout the 30 years with a sudden increase after the 2010 election. Due to the timing of this peak, we can infer the president elected in 2010 impacted the discussion surrounding education policies and institutional frameworks.

Topic 18 remained the most constant of all the topics over the course of 30 years, indicating very little changes in social development policies and economic strategies. Topic 15 also stayed at a similar prevalence level until the 2018 election where it significantly increased, hitting its peak in 2020 around the time when the COVID-19 Pandemic started.

Trends are flat between 2014 and 2018 because we do not include that period in our sample. The slope of the lines simply reflect the jump from the topic's prevalence from 2014 to 2018 directly.

5 Limitations

As we reflect on the validity of our approach and inherent limitations, it is necessary to recognize our methodology approach and the challenges that we encountered. One limitation that was evident from the start included the use of synthesized versions of bills from the Brazilian congress API. By using synthesized versions of bills it potentially omitted details that may have been present in full bill texts thus emphasizing the need for caution when generalizing our findings. Furthermore, by filtering out bills proposed by the executive, party leaderships, and civil society it could have inadvertently skewed the patterns that we observed.

Additionally, when making the decision for the topic labels we were faced with the difficult decision to run a 50 iteration model instead of letting it fully converge. Due to time constraints and the high volume of data it was significantly harder to reach similar results. This decision was a limitation to the research since it took an extensive amount of time and the workflow became impossible between three people to work together. Despite these constraints, our findings managed to provide valuable insights and allowed us to recognize such limitations in order to approach our conclusions with a nuanced understanding of the research. Ultimately, by addressing these limitations we may be able to refine our methodological approach to enhance the research in the future.

6 Appendix

Topic 1 Top Words: Highest Prob: comissão, especí, direito, defesa, criação, destinada, humano FREX: humano, subcomissão, minoria, proferir, sere, redacao, estudar Lift: “grupo, “manifesto, “pandemia, antecédida, assimetria, atípica, baháí Score: comissão, especí, direito, humano, subcomissão, destinada, criação Topic 2 Top Words: Highest Prob: pública, realização, audiência, requer, debat, discutir, sobr FREX: discutir, debat, realização, audiência, expositor, copa, cidades-sed Lift: “abuso, “atualização, “debat, “década, “desafio, “incorporação, “medida Score: audiência, realização, requer, debat, pública, discutir, seminário Topic 3 Top Words: Highest Prob: dia, emenda, inclusão, ordem, requer, constituição, proposta FREX: pronunciamento, diminuição, cabendo, ordem, implicação, aglutinativa, dia Lift: aglutina, aposentar-, “considera, “sobr, acupunturamtc, aglutinativa, aposentar Score: emenda, dia, ordem, inclusão, requer, proposta, pec Topic 4 Top Words: Highest Prob: nova, mpv, anexo, meta, estratégia, seguint, modifique- FREX: nova, navios-escolta, modifique-, modiffique-, estratégia, meta, anexo Lift: modiffique-, “paixão, cidade-teatro, jerusalém,” navios-escolta, revisando-, redaçã Score: nova, mpv, anexo, meta, estratégia, modifique-, acrescente- Topic 5 Top Words: Highest Prob: deputado, câmara, requer, termo, interno, regimento, sessão FREX: deputado, câmara, regimento, sessão, solen, homenagem, ricd Lift: combinado, dign, regimento, requeremo, ricd, sessão, venho Score: deputado, solen, câmara, regimento, sessão, requer, interno Topic 6 Top Words: Highest Prob: requer, parlamentar, estado, justiça, relação, segurança, denúncia FREX: moção, repúdio, aplauso, pazuello, ataque, haiti, gsi Lift: “atrapalham,” “movimento, “orçamento, “reservado,” acompanhará, adir, admar Score: requer, desarquivamento, parlamentar, moção, coautoria, proposição, frent Topic 7 Top Words: Highest Prob: art, medida, provisória, modifica, present, suprima-, dada FREX: provisória, suprimir, present, suprima-, altere-, dispositivo, subprojetosubatividad Lift: “p,” §ºdo, bi-tributação, corretagen, deslocálo, excetua, incua- Score: art, provisória, medida, suprima-, modifica, present, caput Topic 8 Top Words: Highest Prob: educação, ensino, superior, todo, ano, instituição, rede FREX: ensino, estudant, fundament, currículo, curricular,

curso, médio Lift: alfabetização, -licenciado, -luz, -negro, -suprida, “base, “contribuição
Score: educação, ensino, superior, instituição, rede, médio, básica Topic 9 Top Words: Highest
Prob: requerimento, retirada, pauta, matéria, despacho, votação, requer FREX: requeri-
mento, despacho, aditamento, aposto, nomin, req, adiamento Lift: -cft, -cme, ‘combinado,’
“transparência, apyãwa, caucaiac, consagrada Score: requerimento, retirada, pauta, despacho,
matéria, votação, revisão Topic 10 Top Words: Highest Prob: artigo, acrescenta, inciso,
constituição, parágrafo, seguint, art FREX: parágrafo, transitória, inclua-, couber, paragrafo,
cessionário, cedent Lift: “inciso, articulará, atribuir-, coordenará, daproposta, parágrado,
pontilhada Score: artigo, acrescenta, inciso, parágrafo, constituição, seguint, art Topic 11
Top Words: Highest Prob: dep, parec, substitutivo, legislativa, relat, técnica, constituionali-
dad FREX: constitucionalidad, juridicidad, proferido, conclui, subemenda, psdb-sp, dem-sp
Lift: caleffi, carolin, cherini, cidadania-mg, cidadania-sp, er-, fahrur Score: dep, juridicidad,
constitucionalidad, relat, parec, substitutivo, rejeição Topic 12 Top Words: Highest Prob:
redação, mpv, provisória, alaltera, liii, dáa, psdbgo FREX: alaltera, redação, liii, niova, mpv,
dáa, adaptálo Lift: niova, alaltera, liii, adaptálo, autografo, redação, dáa Score: redação,
mpv, provisória, alaltera, liii, dáa, niova Topic 13 Top Words: Highest Prob: energia, meio,
ambient, mina, agricultura, nacion, control FREX: aneel, renovávei, agraria, colonização,
suden, aquicultura, ambient Lift: aden, aluminio, amêndoa, belotti, beriz, boipeba, borra
Score: energia, ambient, agricultura, meio, mina, abastecimento, elétrica Topic 14 Top Words:
Highest Prob: transport, governo, cidad, brasil, estado, república, obra FREX: rodovia, br-,
trecho, federativa, extraordinário, denomina, abr Lift: “acordo, “anõ, “mensalão,” “trevo,
abeba, abr, abrangem Score: cidad, governo, br-, rodovia, república, transport, federativa
Topic 15 Top Words: Highest Prob: saúd, pública, covid-, atividade, durant, trabalhador,
sobr FREX: covid-, pandemia, coronavírus, calamidad, vacinação, enquanto, reconhecido
Lift: enquanto, qualitativa, corona, decretado, durar, durarem, epidemia Score: saúd, covid-,
pandemia, coronavírus, emergência, calamidad, pública Topic 16 Top Words: Highest Prob:
civil, processo, casa, república, presidência, procedimento, caso FREX: aviação, judiciais,

ministra-chef, julgamento, licitatório, aeroportuária, reparação Lift: “caserna, acarretará, alegado, altcoin, anônima, anticres, apelação Score: civil, processo, presidência, república, casa, aviação, procedimento Topic 17 Top Words: Highest Prob: sobr, providência, outra, dispõ, nacion, institui, cria FREX: munção, sinarm, incorporação, outra, cria, fogo, defin Lift: -dgpf, “intimidação, “sinal, absolutória, anorexia, apicun, armas-sinarm Score: outra, dispõ, providência, sobr, institui, cria, nacion Topic 18 Top Words: Highest Prob: social, desenvolvimento, nacion, tecnologia, previdência, combat, indústria FREX: comércio, organizado, inovação, ciência, climática, econômico, tecnologia Lift: “vaccin, adesobra, ahf, alavanca, ampliaria, assistência, bissexuai Score: desenvolvimento, social, tecnologia, indústria, ciência, comércio, nacion Topic 19 Top Words: Highest Prob: president, convidado, esclarecimento, prestar, senhor, requer, voto FREX: pesar, srs, congratulação, comparecerem, evangelização, convida, depor Lift: abal, abeta, abef, abepet, abiov, abipec, abiquim Score: convidado, president, prestar, esclarecimento, louvor, senhor, requer Topic 20 Top Words: Highest Prob: sobr, pessoa, imposto, produto, uso, dispõ, renda FREX: imposto, industrializado, ipi, bebida, microempresa, mercadoria, lucro Lift: fumífero, irpf, -brisa, -choqu, -entrega, -fumant, -ionizant Score: imposto, sobr, produto, pessoa, dispõ, veículo, obrigatoriedad Topic 21 Top Words: Highest Prob: lei, altera, dezembro, julho, setembro, código, dispositivo FREX: dispor, trânsito, permitir, prever, altera, contravenção, auxílio-inclusão Lift: remição, “idoso,” “idosos,” “ncov-,” “pessoa, “regula, ´rgao Score: altera, lei, dezembro, código, julho, dispor, decreto-lei Topic 22 Top Words: Highest Prob: feder, cargo, militar, distrito, polícia, federa, carreira FREX: carreira, bombeiro, plebiscito, distrito, auditoria-fisc, militar, gratificação Lift: hfa, abcdmrr, abrigarem, acautelamento, adidância, advogar, aliment Score: feder, cargo, carreira, polícia, militar, distrito, autoriza Topic 23 Top Words: Highest Prob: suger, município, estado, ministério, implantação, ministro, rio FREX: estiagem, impacta, samu, aed, aegypti, drenagem, uom Lift: -desativação, afuápa, alferesrj, altiplano, alto-rj, anatólia, aperibérj Score: suger, município, implantação, estado, ministro, rio, ministério Topic 24 Top Words: Highest Prob: parec, aprovação, dep, relat, marco, pereira, aprovado FREX:

republic-sp, pl-am, prb-sp, pl-mg, portela, lincoln, psd-am Lift: erika, pdt-rj, pl-pr, pt-pi, aliel, avante-ba, bomfim Score: dep, parec, relat, aprovação, republic-sp, pl-am, pereira Topic 25 Top Words: Highest Prob: informação, solicita, sobr, ministro, estado, acerca, senhor FREX: proger, informação, consignada, mantega, verba, técnicofinanceiro, aeronautica Lift: -liberação, -orçamentario, “assinante,” “cidad, “cuidado, “varíola, acerto Score: informação, solicita, ministro, sobr, ministerio, estado, senhor Topic 26 Top Words: Highest Prob: projeto, lei, requer, complementar, solicita, legislativo, conjunta FREX: apensação, desapensação, reconstituição, apensamento, desapensamento, desapensado, cabotagem Lift: reconstituição, -correlata, “ministério, “novembrinho, “poder, alteradora, alternando Score: projeto, lei, requer, complementar, apensação, desapensação, apensado Topic 27 Top Words: Highest Prob: público, serviço, trabalho, administração, pública, gestão, entidad FREX: administração, concurso, ps-gse, encaminha, processado, indireta, autarquia Lift: encaminha, sanção, abaixo-assinado, administr, administração, administrativ, adotarão Score: público, administração, ps-gse, gestão, planejamento, serviço, trabalho Topic 28 Top Words: Highest Prob: poder, executivo, ministério, suger, programa, inclusão, município FREX: dentro, aceleração, priorizaçõ, apa, alud, cadastrada, cadastramento Lift: cegonha, analisada, mecfnd, mesorregião, pnte, pólos, priorizaçõ Score: suger, executivo, poder, intermédio, paraná, ministério, inclusão Topic 29 Top Words: Highest Prob: radiodifusão, ato, serviço, estado, comunitária, aprova, executar FREX: radiodifusão, comunitária, executar, exclusividad, dez, renova, outorgada Lift: -ijuí, aacc, aamca, aamepb, abadiânia, abaiara, abaré Score: radiodifusão, comunitária, executar, exclusividad, renova, aprova, outorgada Topic 30 Top Words: Highest Prob: decreto, efeito, susta, aplicação, portaria, resolução, nacion FREX: susta, contran, colegiada, normativa, ldo, elaboração, dotaçõ Lift: -colog, -sctiem, ‘dolar, “atualiza, adenocarcinoma, ambiguidad, americano’ Score: susta, decreto, efeito, portaria, resolução, aplicação, orçamentária

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